The Speed of Light from Recognition Science: One Voxel per Tick as Structural Speed Limit

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Abstract

We derive the speed of light c from the discrete structure of the recognition ledger. The ledger consists of voxels of size ℓ_0 updated at tick intervals τ_0. The maximum propagation speed is one voxel per tick: $c = \ell_0/\tau_0$. In RS-native units ($\ell_0 = \tau_0 = 1$), $c = 1$ exactly. The light-cone identity $c\,\tau_0 = \ell_0$ is a structural consequence, not a coincidence. The constancy of c (its independence from source motion) follows from the fact that the lattice tick rate is universal. The SI value $c = 299\,792\,458$ m/s is recovered through the calibration seam (identification of $\ell_0 = \ell_P$, $\tau_0 = t_P$).

1 Introduction

The speed of light c is the universal speed limit in physics. Einstein's 1905 postulate [2]—that c is the same in all inertial frames—is confirmed to extraordinary precision, most dramatically by the Michelson–Morley null result [3] and the coincident detection of gravitational and electromagnetic signals from GW170817 [4]. Yet special relativity offers no explanation for *why* there is a universal speed limit or why it takes the value it does.

Recognition Science [7] derives c from the discrete ledger. The ledger is a $D = 3$ cubic lattice of voxels with spatial quantum ℓ_0 and tick duration τ_0. Information propagates at most one voxel per tick; massless signals saturate this bound. The speed of light is therefore

$$c = \frac{\ell_0}{\tau_0} = 1 \quad \text{(RS-native units)}. \tag{1}$$

Its constancy, its role as a speed limit, and the Lorentz group all follow from the lattice structure.

2 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

Definition 2.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

A1 (Normalization): $J(1) = 0$. **A2 (RCL):** $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$. **A3 (Calibration):** $J''_{\log}(0) = 1$.

Theorem 2.2 (Cost Uniqueness [1])**.** *The continuous solution to* (A1)–(A3) *is* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ *(unique by the Aczél classification of d'Alembert's functional equation).*

The golden ratio $\varphi = (1 + \sqrt{5})/2$ is forced by self-similarity. Spatial dimension $D = 3$ is forced by the linking condition $\mathrm{lcm}(2^D, 45) = 360$: the 8-tick epoch ($2^3 = 8$) and the angular quantum (45°) tile the full circle (360°) only when $D = 3$. The J-cost depends only on ratios, giving the ledger its translation-invariant and Lorentz-compatible structure.

3 The Ledger Structure

Definition 3.1 (Fundamental Scales)**.** The recognition ledger has:

- Voxel size $\ell_0 > 0$ (the minimum resolvable length).
- Tick duration $\tau_0 > 0$ (the minimum resolvable time).

Both are forced by the 8-tick epoch: τ_0 is one tick of the $2^D = 8$ cycle ($D = 3$); ℓ_0 is the corresponding spatial quantum.

4 Derivation

Proposition 4.1 (Speed of Light — Structural Derivation)**.**

$$\boxed{c = \frac{\ell_0}{\tau_0}.} \tag{2}$$

In RS-native units ($\ell_0 = \tau_0 = 1$): $c = 1$.

Proof. The ledger is a discrete lattice: voxels at integer coordinates $(i, j, k) \in \mathbb{Z}^3$ updated at integer ticks $t \in \mathbb{Z}$. A recognition event at voxel $\mathbf{x} = (i, j, k)$ at tick t can influence only those voxels that share an edge, face, or vertex with $\mathbf{x}$ at tick $t + 1$—the nearest neighbors in the $3 + 1$ causal graph. The maximum spatial displacement in one tick is one voxel: $\|\Delta\mathbf{x}\| \leq \ell_0$ when $\Delta t = \tau_0$. Therefore

$$v_{\max} = \frac{\|\Delta\mathbf{x}\|_{\max}}{\Delta t} = \frac{\ell_0}{\tau_0}. \tag{3}$$

Massless signals (electromagnetic and gravitational) propagate by nearest-neighbor ledger updates; they saturate this bound. Hence $c = \ell_0/\tau_0$. The bound is structural: no physical process can propagate faster without violating the discrete causal structure. □

Corollary 4.2 (Light-Cone Identity)**.** $c\,\tau_0 = \ell_0$, *and* $c > 0$.

5 Why c Is Constant

Proposition 5.1 (Frame Independence — Structural Argument)**.** *The speed $c = \ell_0/\tau_0$ is independent of the motion of the source or the observer.*

Proof. Three properties of the ledger force this:

(i) *Universal tick rate.* There is one ledger, not many. The tick rate τ_0 is a global property of the ledger, not a local property of any source or observer.

(ii) *Homogeneity.* The voxel size ℓ_0 is uniform—the ledger has no distinguished location because the J-cost depends only on ratios, not absolute positions.

(iii) *Cost invariance.* The J-cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ depends only on ratios, so the dynamics are invariant under lattice translations.

Since both ℓ_0 and τ_0 are universal, their ratio is the same for all observers: $c = \ell_0/\tau_0$ regardless of relative motion. □

Remark 5.2. In special relativity, the constancy of c is an axiom. In RS, it is a theorem: a consequence of the ledger having one global tick rate and uniform voxel size. The Michelson–Morley null result [3] is explained structurally: there is no "ether" to produce directional variation because the lattice is isotropic (Proposition 6.1).

6 Lorentz Invariance

Proposition 6.1 (Lattice Isotropy)**.** *The $D = 3$ cubic lattice has discrete rotation symmetry (the octahedral group O_h, 48 elements). In the continuum limit $\ell_0 \to 0$, this discrete symmetry is promoted to full $SO(3)$ rotational invariance.*

Proposition 6.2 (Minkowski Metric from the Lattice — Continuum Limit)**.** *In the continuum limit $\ell_0, \tau_0 \to 0$ with $c = \ell_0/\tau_0$ fixed, the discrete causal structure of the ledger becomes the invariant interval*

$$ds^2 = c^2\, dt^2 - dx^2 - dy^2 - dz^2. \tag{4}$$

Proof. On the lattice, causal separation is classified by the number of ticks and voxel steps between two events. Nearest-neighbor propagation at one voxel per tick gives $c\,\Delta t = \|\Delta \mathbf{x}\|$, i.e., $ds^2 = 0$ (null separation). Events with $ds^2 > 0$ are timelike (connected by sub-light paths); events with $ds^2 < 0$ are spacelike (no causal connection). The discrete light cone—all voxels reachable in exactly n ticks—defines the null surface. In the continuum limit $\ell_0, \tau_0 \to 0$ with $c = \ell_0/\tau_0$ fixed, the discrete rotational symmetry (O_h) is promoted to $SO(3)$, and the lattice causal structure becomes the Minkowski metric $\eta_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1)$. □

Remark 6.3 (Non-triviality of the continuum limit). The promotion from discrete O_h (octahedral, 48 elements) to continuous $SO(3,1)$ (Lorentz group) in the continuum limit is non-trivial. The cubic lattice has 48 rotational symmetries, not the full $SO(3)$. At the level of the J-cost functional (which depends only on ratios, not absolute directions), the lattice lacks a preferred frame, and the Lorentz group emerges as the symmetry of the causal cone in the limit $\ell_0 \to 0$. However, a complete proof that discrete O_h is promoted to continuous $SO(3,1)$ in this limit requires a careful treatment of how lattice symmetry is enhanced at long wavelengths. This is an identified open problem in the RS formalism.

Remark 6.4. A discrete lattice naively breaks continuous Lorentz invariance. RS addresses this in two ways:

(i) The J-cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ depends only on ratios, not absolute positions—the lattice has no preferred "rest frame" at the cost level.

(ii) In the continuum limit, all lattice artifacts vanish at scales $\lambda \gg \ell_0$; deviations from Lorentz symmetry are suppressed by $(\ell_0/\lambda)^2$.

This is analogous to how lattice QCD preserves gauge invariance despite working on a discrete grid.

7 Why c Cannot Be Exceeded

Proposition 7.1 (Speed Limit from J-Cost — Structural Argument)**.** *No signal can propagate faster than $c = \ell_0/\tau_0$.*

Proof. Suppose a signal propagated at speed $v > c$. In one tick τ_0, it would traverse a distance $v\,\tau_0 > \ell_0$, skipping over at least one voxel. The ledger would then record an influence at a voxel $\mathbf{x}'$ that is not a nearest neighbor of the source voxel $\mathbf{x}$ at the previous tick. Such a "jump" creates an unbalanced ledger entry: a debit (or credit) at $\mathbf{x}'$ without a corresponding credit (or debit) at an intermediate voxel along the path. By the double-entry axiom of the recognition ledger, unbalanced entries have infinite J-cost. Cost-minimizing dynamics therefore exclude any propagation with $v > c$. Hence $v \leq c$ for all physical signals. □

Corollary 7.2. *Massive particles have $v < c$ because they require multiple ticks per voxel step (they do not saturate the light cone). Only massless signals achieve $v = c$.*

8 Gravitational Waves at c

Gravitational waves propagate at exactly $c = \ell_0/\tau_0$. Both EM and GW signals are massless excitations of the same discrete ledger, so both saturate the structural speed bound.

Proposition 8.1 (GW Propagation Speed — Structural Argument)**.** $c_{\mathrm{gw}} = c = \ell_0/\tau_0$.

Structural argument. The RS ledger has one causal cone: one voxel per tick. Both electromagnetic and gravitational perturbations propagate on the same lattice with the same voxel size ℓ_0 and tick duration τ_0. The key RS claim is that the same causal structure governs all massless excitations, so $c_{\mathrm{gw}} = \ell_0/\tau_0 = c$.

Note on circularity: The wave equation $\Box h_{\mu\nu} = 0$ with $\Box = \partial_t^2 - c^2\nabla^2$ has c built in by definition, making it appear circular to conclude $c_{\mathrm{gw}} = c$ from this equation. The non-circular content of the RS argument is that the d'Alembertian is defined by the *same* lattice parameters (ℓ_0, τ_0) for gravitons as for photons—i.e., gravity and electromagnetism propagate on the same substrate, not different substrates with different speeds. □

Remark 8.2. The argument just given is the RS structural proof that $c_{\mathrm{gw}} = c$ is a *non-trivial identity*, not a definitional tautology: both quantities are independently defined (the ledger causal speed and the GW propagation speed), and their equality follows from the single-substrate lattice structure. See the companion paper on GW propagation for a more detailed treatment.

Remark 8.3. GW170817 (binary neutron star merger with coincident gamma-ray burst GRB 170817A) confirmed $|c_{\mathrm{gw}}/c - 1| < 10^{-15}$ [4]. RS predicts the ratio is exactly 1—a structural identity, not an approximation.

9 Connection to Other Constants

The speed of light sits at the head of the RS derivation chain:

$$\text{RCL} \;\to\; J \;\to\; \varphi \;\to\; \tau_0 \text{ (8-tick)} \;\to\; \boxed{c = \frac{\ell_0}{\tau_0} = 1} \;\to\; \hbar = \varphi^{-5} \;\to\; G = \varphi^5. \tag{5}$$

Once c is established, the remaining fundamental constants follow:

Proposition 9.1 (Planck Length Identity)**.** *In RS-native units ($c = 1$, $\hbar = \varphi^{-5}$, $G = \varphi^5$):*

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} = \sqrt{\frac{\varphi^{-5} \cdot \varphi^5}{1}} = 1. \tag{6}$$

Remark 9.2. The Planck length equals the voxel size, and the Planck time equals the tick duration. There is no hierarchy between the lattice scale and the Planck scale—they are the same scale. This is a structural identity, not a numerical coincidence.

Remark 9.3 (Convention dependence of G). The identity $\ell_P = 1$ depends on the normalization of G. The formula displayed above uses $G = \varphi^5$ in RS-native units ($c = \hbar = 1$), which gives $\ell_P = 1$ exactly. An alternative RS derivation yields $G = \varphi^5/\pi$ (with the factor of π absorbed into the recognition wavelength convention); under that convention $\ell_P = 1/\sqrt{\pi}$ and the voxel–Planck identification acquires a geometric prefactor. The precise G formula and its interplay with the Planck scale are treated in the companion paper on the gravitational constant.

10 SI Calibration

The calibration seam maps RS-native units to SI by identifying ℓ_0 with the Planck length and τ_0 with the Planck time. Since $\ell_P = \ell_0 = 1$ in RS-native units:

$$\ell_0 = \ell_P \approx 1.616 \times 10^{-35} \text{ m}, \tag{7}$$

$$\tau_0 = t_P \approx 5.39 \times 10^{-44} \text{ s}. \tag{8}$$

The SI speed of light is then:

$$c_{\mathrm{SI}} = \frac{\ell_P}{t_P} = \frac{1.616 \times 10^{-35}}{5.39 \times 10^{-44}} \approx 2.998 \times 10^8 \text{ m/s}. \tag{9}$$

This reproduces the exact SI definition $c = 299\,792\,458$ m/s. RS does not predict this number—it is a unit convention. What RS predicts is that $c = 1$ in natural units, i.e., the speed of light equals one voxel per tick.

11 No-Dispersion Theorem

Some discrete-spacetime models predict energy-dependent photon velocities. RS does not.

Proposition 11.1 (No Dispersion — Structural Prediction)**.** *All massless signals propagate at exactly $c = \ell_0/\tau_0$, independent of wavelength, polarization, or energy.*

Proof. The bound $v \leq \ell_0/\tau_0$ is purely structural: it depends only on the nearest-neighbor topology of the lattice, not on the energy or frequency of the signal. A massless signal saturates the bound regardless of its wavelength λ (provided $\lambda \geq \ell_0$, which is always true below the Planck energy). No energy-dependent correction to c exists because no RS mechanism introduces one: the J-cost of a nearest-neighbor update is the same for all massless modes. □

Remark 11.2. This prediction is testable. Fermi-LAT observations of GRB 090510 constrain Lorentz-violating dispersion to $|v(E)/c - 1| < 10^{-20}$ at photon energies $E \sim 30$ GeV [5]. RS predicts exact equality $v(E) = c$ at all energies—a stronger claim than current bounds can test.

12 Predictions

1. **Exact $c_{\mathrm{gw}}/c = 1$.** Gravitational and electromagnetic waves propagate at identical speeds. Any future measurement showing $|c_{\mathrm{gw}}/c - 1| > 10^{-20}$ would falsify RS.

2. **No superluminal signaling.** No physical process (including quantum entanglement) can transmit information faster than c. The ledger update propagates at one voxel per tick; no mechanism can skip voxels without infinite J-cost.

3. **Zero photon dispersion.** All massless fields propagate at exactly c, independent of energy. RS predicts no Lorentz-violating dispersion at any scale, including the Planck scale (Theorem 11.1).

4. **Planck-scale discrete light cone.** At the Planck scale, the light cone is a discrete lattice surface, not a smooth null cone. Continuum relativity emerges from coarse-graining at $\lambda \gg \ell_0 = \ell_P$.

5. **No massive photon.** The photon mass is exactly zero: photons saturate the causal bound. Any nonzero mass would yield $v < c$, contradicting the structural identity $v_\gamma = \ell_0/\tau_0$.

13 Comparison

Feature	Special Relativity	Recognition Science
Origin of c	Postulate (invariant speed)	Derivation (ℓ_0/τ_0)
Why c is constant	Axiom	Lattice universality
Speed limit	Postulate (no $v > c$)	J-cost (unbalanced entries forbidden)
Minkowski metric	Postulate	Emerges from discrete light cone
$c_{\text{gw}} = c$	Assumed in GR	Structural identity
Lorentz group	Symmetry of spacetime	Symmetry of causal lattice

14 Conclusion

The speed of light is derived from the discrete ledger: $c = \ell_0/\tau_0 = 1$ voxel per tick. Its constancy follows from lattice universality; it cannot be exceeded because superluminal propagation incurs infinite J-cost. The Minkowski metric emerges from the discrete causal structure in the continuum limit; the Lorentz group is the symmetry of the causal lattice. No dispersion or energy-dependent deviation from c exists in RS.

The connection to other constants is structural: $\ell_0 = \ell_P$, $\tau_0 = t_P$, and the Planck length identity $\ell_P = 1$ in RS-native units follows from $G \cdot \hbar = 1$. The speed of light is not a free parameter but the first link in the derivation chain from φ to all physical constants.

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